Yuan Jack Yao

404-528-6519 | yyao411@gatech.edu | linkedin.com/in/yuan-jack-yao | starsexpress.github.io/Faraway-s-Way

Education

Georgia Institute of Technology **Aug 2025 – Dec 2026**
M.S. in Analytics @ ***School of Computational Science and Engineering*** *Atlanta, Georgia*

- CS/CSE Courses: Conversational AI, Natural Language, Machine Learning, ML in Computational Biology

University of California, San Diego **Sep 2019 – Sep 2020**
Master of Finance *La Jolla, California*

National Taiwan University **Sep 2014 – Jun 2018**
Bachelor's Degree in Economics *Taipei, Taiwan*

Work Experiences

Georgia Institute of Technology **Aug 2026 – Dec 2026**
CS 6604 Conversational AI Graduate Teaching Assistant *Atlanta, Georgia*

- **Deployed** a **Voice Agent** assignment for an 80-student course: EOS→ASR→LLM→TTS pipeline with **4% ASR word error rate** and EOS-to-first-audio **p50 ≤ 275 ms, p95 ≤ 417 ms** across four LLMs vs. 800 ms benchmark
- Designed **RAG with cross-session memory** and **LoRA SFT + DPO** (unit-test autograder) assignments; debugged **AI eval harness** so auto evals separate strong from weak submissions; held weekly office hours for student debugging

Walmart **Jun 2026 – Aug 2026**
Data Scientist III Intern @ Sam's Club AI Acceleration Team *Bentonville, Arkansas*

- Integrated **LLM & RAG with filtered vector search** into a dashboard for camera/area/site-level metadata queries, guarding against hallucinations with **cosine similarity & LLMs-as-a-Judge**; an **embedding signature** verifies Milvus ingestion & retrieval use the identical embedding model, **eliminating silent retrieval degradation**
- Architected a **Cassandra-sourced** data layer feeding **PostgreSQL** and **Milvus**, regenerable after data loss
- Deployed Docker & Kubernetes CronJobs for **1,245 in-store cameras across 9 sites**: monthly Gemini Vision area classification of the **full fleet in 13 minutes** via high-throughput concurrency, with glibc malloc arena caps and image pre-downsampling **eliminating OOM failures**
- Optimized hourly **asynchronous snapshot capture** to Azure, cutting **50% time** and **75% storage cost** via 7-day TTL

SUEZ **May 2022 – Sep 2024**
Junior Data Scientist @ Smart and Environmental Solutions (SES) China *Shanghai, China*

- Streamlined **NLP classifier** for **client's imbalanced** SCADA alarms, reaching **82% production accuracy**
- Designed a **DBSCAN anomaly detector** for **client's** air compressors predictive maintenance, achieving **70% production F1 score** and forecasting failures **two weeks in advance despite having only a single feature**

Personal Projects & Products

LinesShines (www.lines-shines.com): NFL & NCAAF Stats System, **iOS/Android/macOS/Windows Installable**

- Solely owned an installable NFL & NCAAF metrics PWA end to end: preprocessing pipeline, PostgreSQL ingestion, FastAPI JSON API, vanilla-JS frontend with Service Worker caching, 560+ pytest/Playwright tests, deployed on Railway behind CI
- Designed pinnable multi-merging player comparison system with live-recomputing rank & percentile panels across multi-team/player union filters, inspired by GitHub's pinned repos & pull requests merge mechanics
- Engineered AI-assisted workflows with enforced guardrails: Claude Code hooks (block restricted commands, keep licensed data out of repo, test + lint gates), layered docs cutting the always-loaded instructions from 79 KB to under 3 KB, and CI that fails when a test lacks its scope doc

Open Source

Pydantic AI (21.1k Stars, 3.1k Forks) — Agentic AI Framework — github.com/pydantic/pydantic-ai

- **Unified API-error handling** across Google & Cohere embedding models with shared context managers

Code Puppy (880 Stars, 297 Forks) — Coding Agents — github.com/mpfaffenberger/code_puppy

- **12+ merged PRs: cut down idle CPU wake-ups by 88.5% for Agent-UI communications, restored silently-dropped user config across Gemini classes**, unified tools' drifted file-read process, and aligned 20+ parser calls

LeetCode Wiki (36.8k Stars, 9.5k Forks) — Data Structures & Algorithms — github.com/doocs/leetcode

- Authored **8 optimal solutions with tutorials** for **hard** problems, including No.2454 in O(n) via double stacks

Technical Skills

Agents & ML: AI Harness & Evals, ASR, TTS, RAG, Distillation, LoRA SFT, DPO, PyTorch, Hugging Face, TensorFlow, Keras
Programming & Databases: Python, SQL, C++, JavaScript, PostgreSQL, MySQL, Milvus, Cassandra, ETL
Web & Backend: FastAPI, Flask, REST APIs, OpenAPI, HTML, CSS, Plotly, Web Scraping
CI/CD, Tests & Cloud: CI/CD, pytest, Playwright, Linux, GitHub, Docker, Kubernetes, Azure, AWS, GCP